UNIVERSITY ERP SYSTEM PROJECT REPORT

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Introduction

This project implements a desktop-based University ERP System using Java + Swing, fulfilling all specifications.

The system supports three types of users:

- Student – register/drop sections, view timetable & grades, download transcript.
- Instructor – manage assigned sections, enter scores, compute final grades, view stats.
- Admin – user management, course/section creation, instructor assignment, maintenance mode.

Data is stored in two separate databases for security:

1. Auth DB – username, role, password hash, last login.
2. ERP DB – students, instructors, courses, sections, enrollments, grades, system settings.

This separation ensures password safety and aligns with the UNIX "shadow file" approach

System Architecture

The project uses a layered & modular architecture recommended by the brief:

a. UI Layer (Swing)

Contains all screens grouped by role (Student, Instructor, Admin).

b. Domain Layer

Simple Java classes:
Student, Instructor, Course, Section, Enrollment, Grade, Settings.

c. Service Layer

Handles all operations like:

- register/drop
- grade entry
- maintenance mode checks
- validation enforcement

d. Data Access Layer

JDBC queries for both AuthDB & ERPDB.

e. Authentication Package

1. Password hashing (bcrypt)
2. Role verification
3. Session management

f. Utility Layer

1. CSV export
2. Validation
3. Messaging

Access Control & Maintenance Rules

Role-based permissions

- Admin → full access.
- Instructor → only their own sections & grade entries.
- Student → only their own registrations & academic records.

Maintenance Mode

When ON:

- Students & instructors → read-only (cannot register/drop or enter grades).
- Admin → full access continues.

A visible yellow banner appears in the UI indicating restricted mode.

Features Implemented

A. Common Features

- Login with password hashing
- Role-based dashboard
- Clean error/validation messages

B. Student Module

- Browse full course catalog
- Register/drop (duplicate & capacity checks included)

- View timetable
- View assessments & final grade
- Download transcript (CSV)

C. Instructor Module

- View assigned sections
- Create assessment components
- Enter scores
- Compute final grade using weighting rule
- View class statistics (average)

D. Admin Module

- Add users (Auth DB)
- Create/manage courses & sections
- Assign instructors
- Toggle maintenance mode
- Manage program/year data

Data Storage Design

Auth DB

users_auth table

- user_id
- username
- role
- password_hash

Password is never stored, only the bcrypt hash.

ERP DB

Tables:

- students
- instructors
- courses
- sections
- enrollments
- grades
- settings (maintenance flag)

Final Grade Calculation

Instructors define a weight scheme (e.g., Quiz 20%, Midterm 30%, Final 50%).
System computes weighted marks and stores the final grade.

Sample Data Provided

- 1 Admin
- More than one Instructor with assigned section
- More than two Students with sample enrollments and grades
- Auth DB contains only hashed passwords

University ERP System — Project...

Testing Summary

✔ Login Tests

Correct role dashboards, wrong password rejection.

✔ Student Tests

- Catalog display
- Register/drop
- Seat capacity check
- Duplicate enrollment prevention
- Transcript export

✔ Instructor Tests

- Edit only their sections
- Enter assessments
- Compute final grade

✔ Admin Tests

- Create student/instructor users
- Create courses/sections
- Toggle maintenance mode

✔ Data Integrity Tests

- No duplicate enrollments
- Safe deletion handling

✔ UI/UX Tests

- Clear labels
- No crashes
- Sorting & validation

Overall almost all tests passed successfully.

Additional Enhancements (Bonus Features)

Based on your implementation:

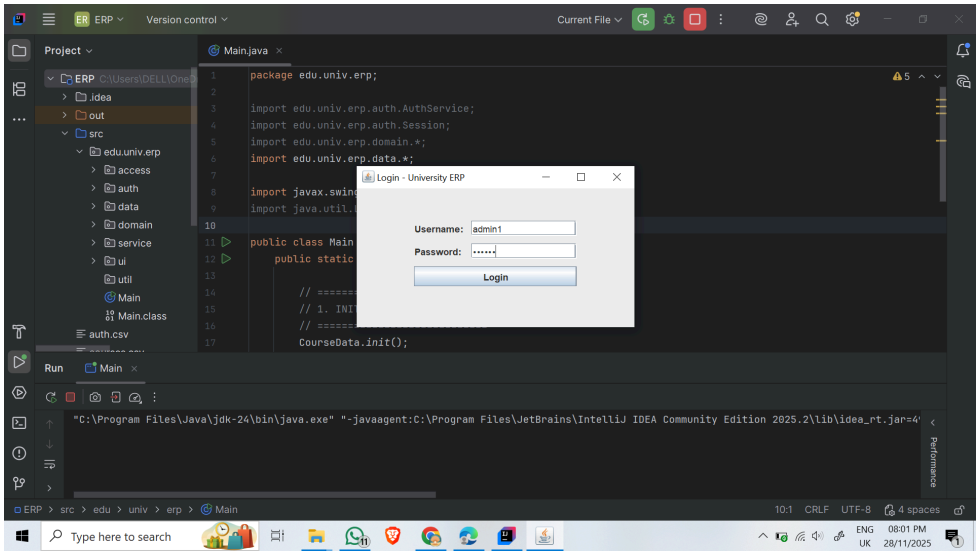
- Login attempt lockout
- Password change
- Notification banner
- Automatic CSV updates

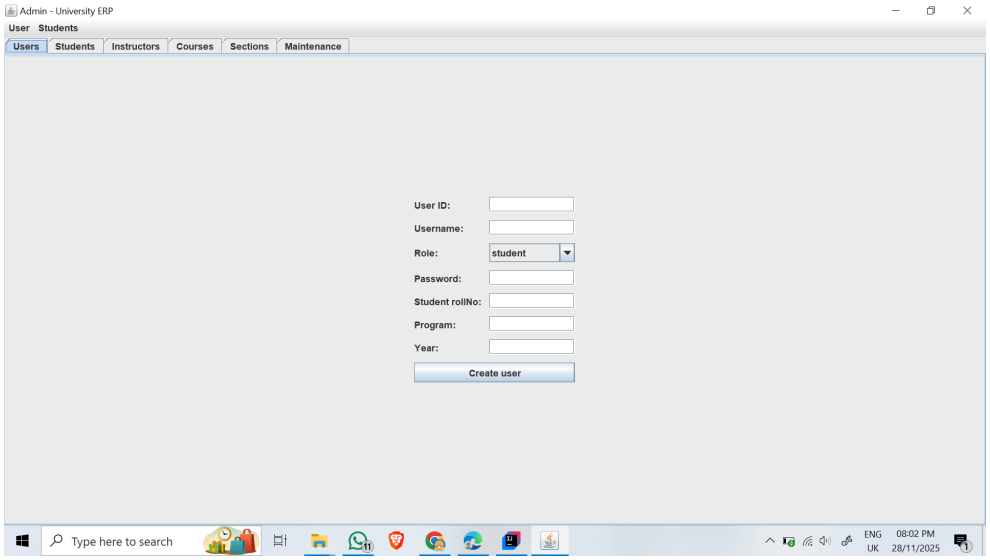
Conclusion

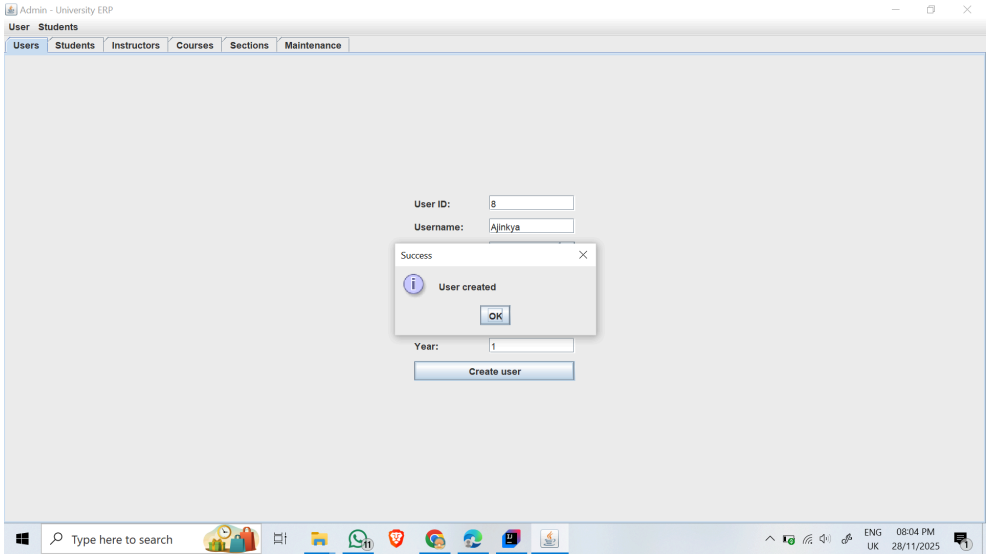
The University ERP System fully meets the requirements:

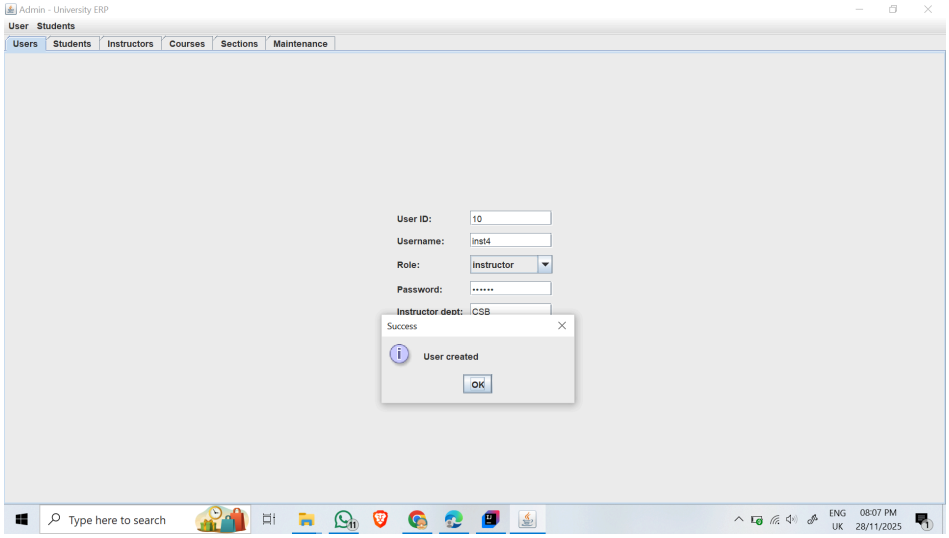
- Secure authentication (two-DB model)
- Clear separation of roles
- Accurate enforcement of the maintenance mode
- Robust course/section/grade management
- Proper validation & error handling
- Clean UI and stable performance

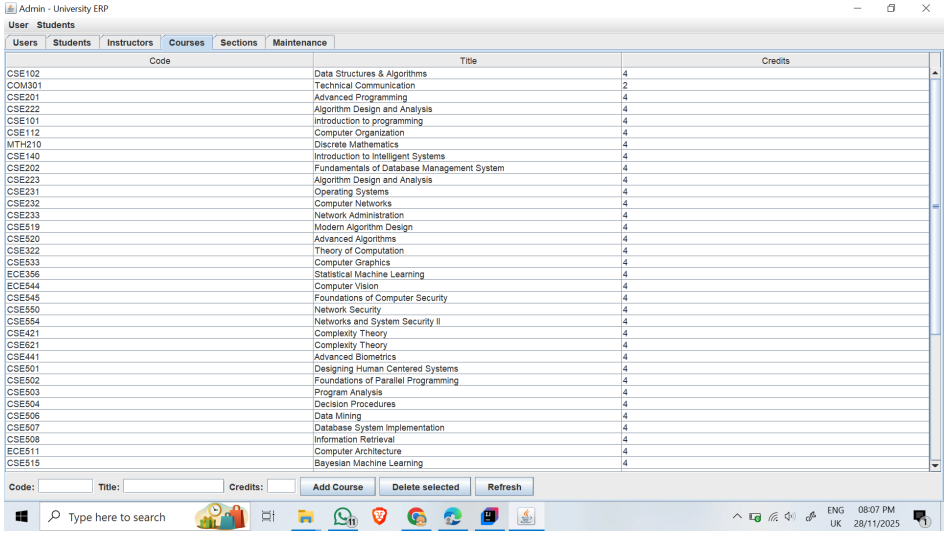
Gallery :



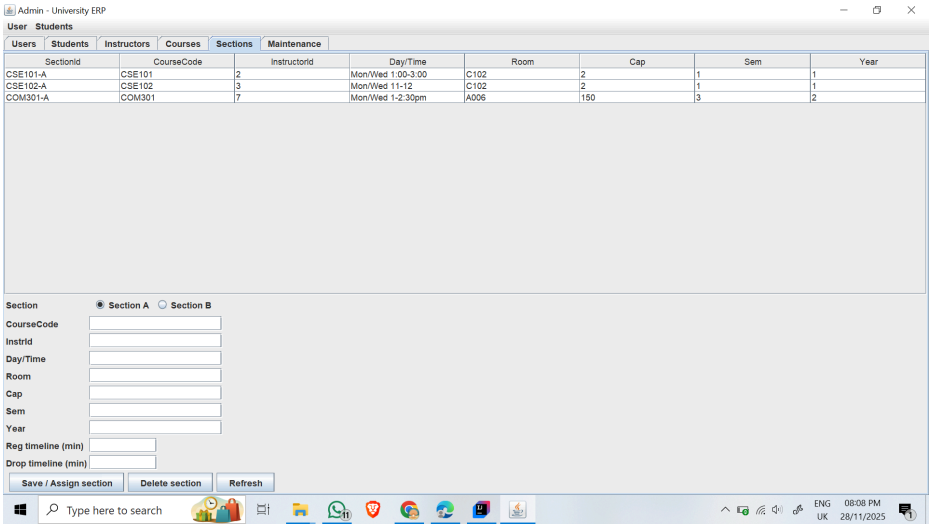

Admin - University ERP
User Students
Users Students Instructors Courses Sections Maintenance
User ID:
Username:
Role:
student
Password:
Student rollNo:
Program:
Year:
Create user
Type here to search
ENG
UK
08:02 PM
28/11/2025

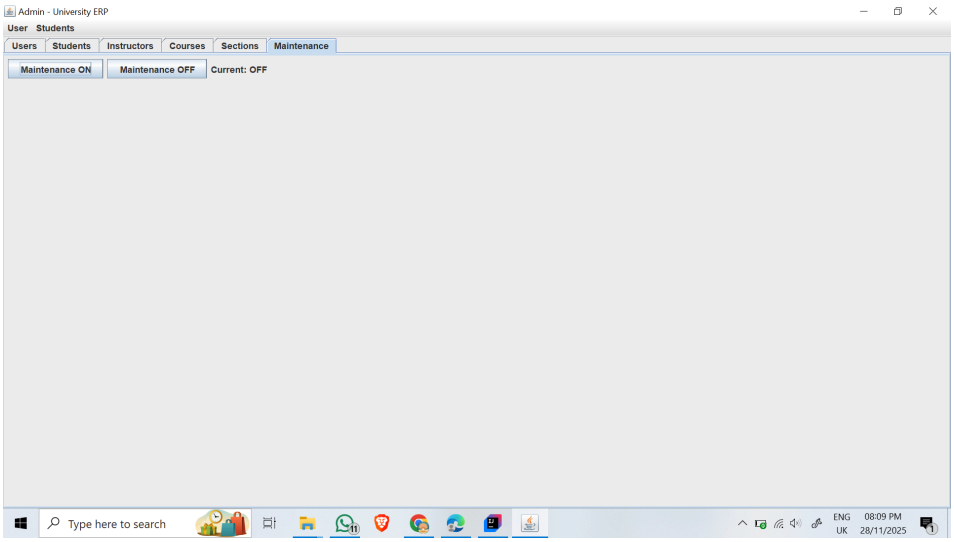

Admin - University ERP
User Students
Users Students Instructors Courses Sections Maintenance
User ID:
8
Username:
Ajinkya
Success
User created
OK
Year:
1
Create user
Type here to search
ENG
UK
08:04 PM
28/11/2025

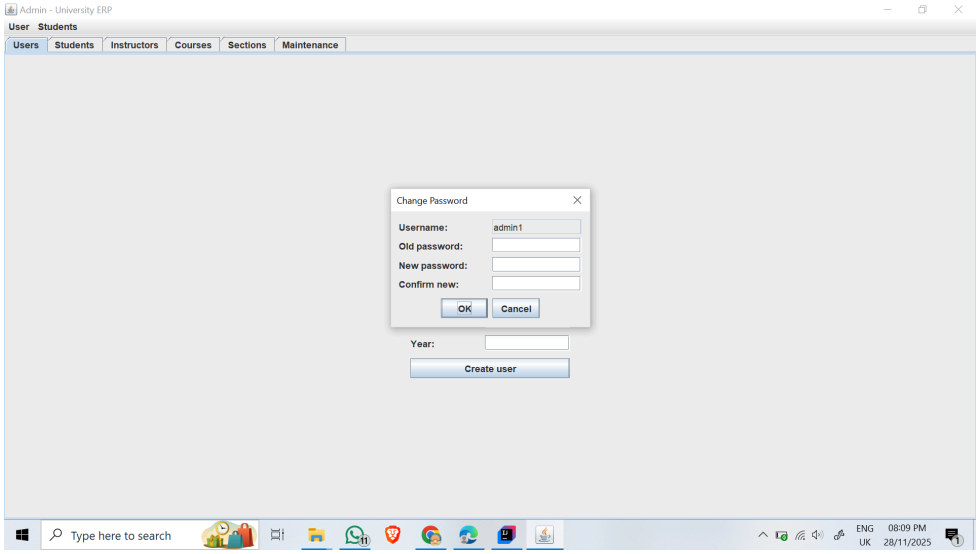


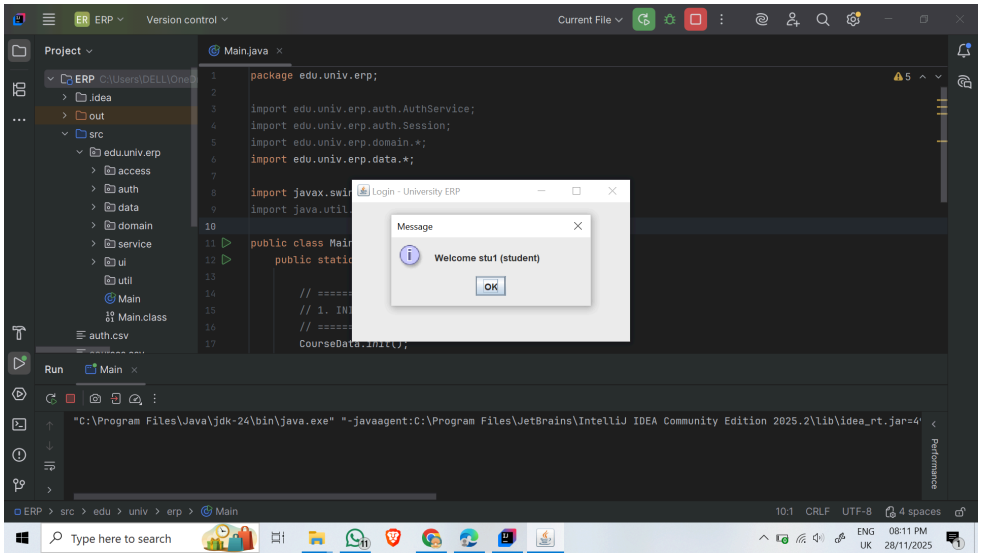


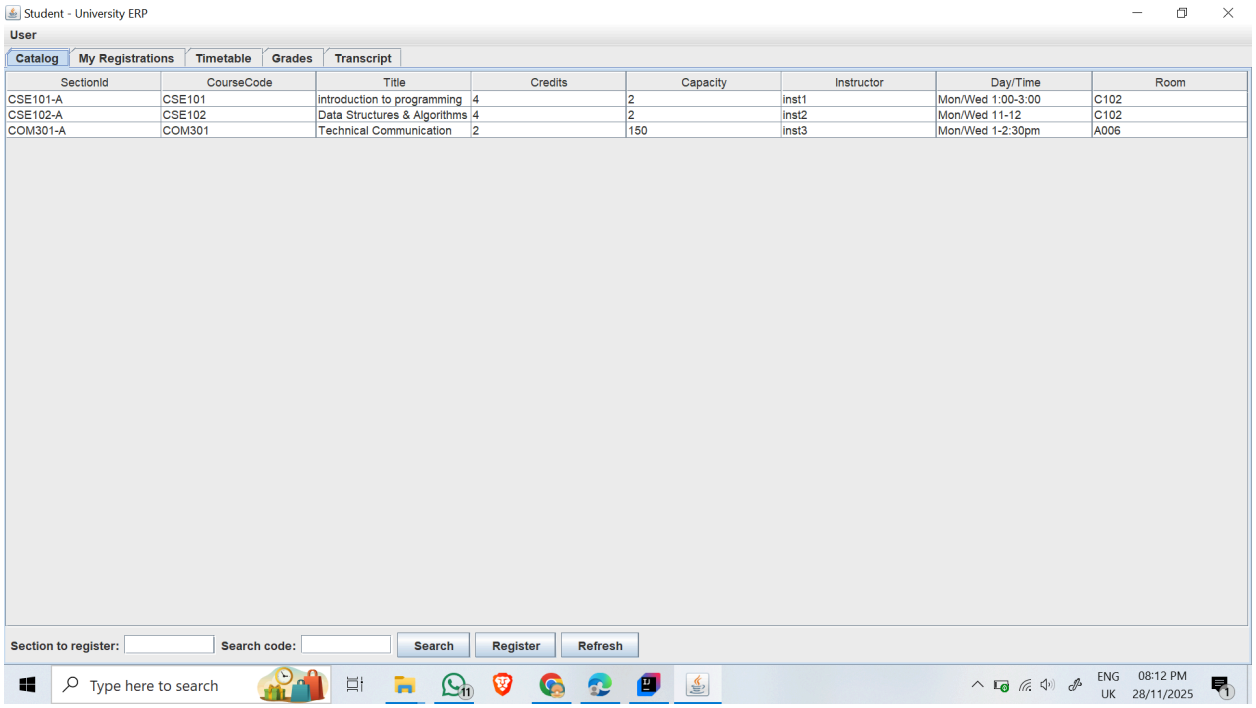
Code	Title	Credits
CSE102	Data Structures & Algorithms	4
COM301	Technical Communication	2
CSE201	Advanced Programming	4
CSE222	Algorithm Design and Analysis	4
CSE101	introduction to programming	4
CSE112	Computer Organization	4
MTH210	Discrete Mathematics	4
CSE140	Introduction to Intelligent Systems	4
CSE202	Fundamentals of Database Management System	4
CSE223	Algorithm Design and Analysis	4
CSE231	Operating Systems	4
CSE232	Computer Networks	4
CSE233	Network Administration	4
CSE519	Modern Algorithm Design	4
CSE520	Advanced Algorithms	4
CSE322	Theory of Computation	4
CSE533	Computer Graphics	4
ECE356	Statistical Machine Learning	4
ECE544	Computer Vision	4
CSE545	Foundations of Computer Security	4
CSE550	Network Security	4
CSE554	Networks and System Security II	4
CSE421	Complexity Theory	4
CSE621	Complexity Theory	4
CSE441	Advanced Biometrics	4
CSE501	Designing Human Centered Systems	4
CSE502	Foundations of Parallel Programming	4
CSE503	Program Analysis	4
CSE504	Decision Procedures	4
CSE506	Data Mining	4
CSE507	Database System Implementation	4
CSE508	Information Retrieval	4
ECE511	Computer Architecture	4
CSE515	Bayesian Machine Learning	4


Admin - University ERP
User Students
Users Students Instructors Courses Sections Maintenance
SectionId CourseCode InstructorId Day/Time Room Cap Sem Year
CSE101-A CSE101 2 Mon/Wed 1:00-3:00 C102 2 1 1
CSE102-A CSE102 3 Mon/Wed 11-12 C102 2 1 1
COM301-A COM301 7 Mon/Wed 1-2:30pm A006 150 3 2
Section Section A Section B
CourseCode
InstrId
Day/Time
Room
Cap
Sem
Year
Reg timeline (min)
Drop timeline (min)
Save / Assign section
Delete section
Refresh
Type here to search
ENG UK 08:08 PM 28/11/2025

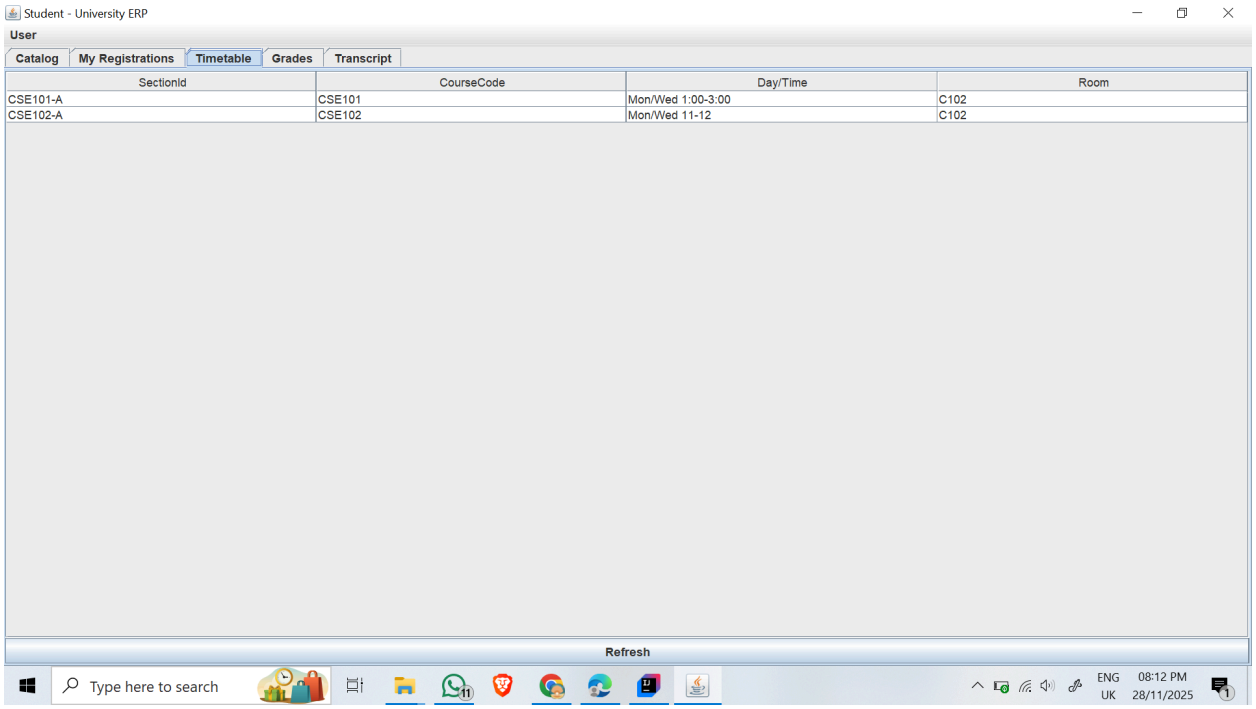

Admin - University ERP
User Students
Users Students Instructors Courses Sections Maintenance
Maintenance ON
Maintenance OFF
Current: OFF
Type here to search
ENG UK 08:09 PM 28/11/2025


Admin - University ERP
User Students
Users Students Instructors Courses Sections Maintenance
Change Password
Username: admin1
Old password:
New password:
Confirm new:
OK Cancel
Year:
Create user
Type here to search
ENG UK 08:09 PM 28/11/2025

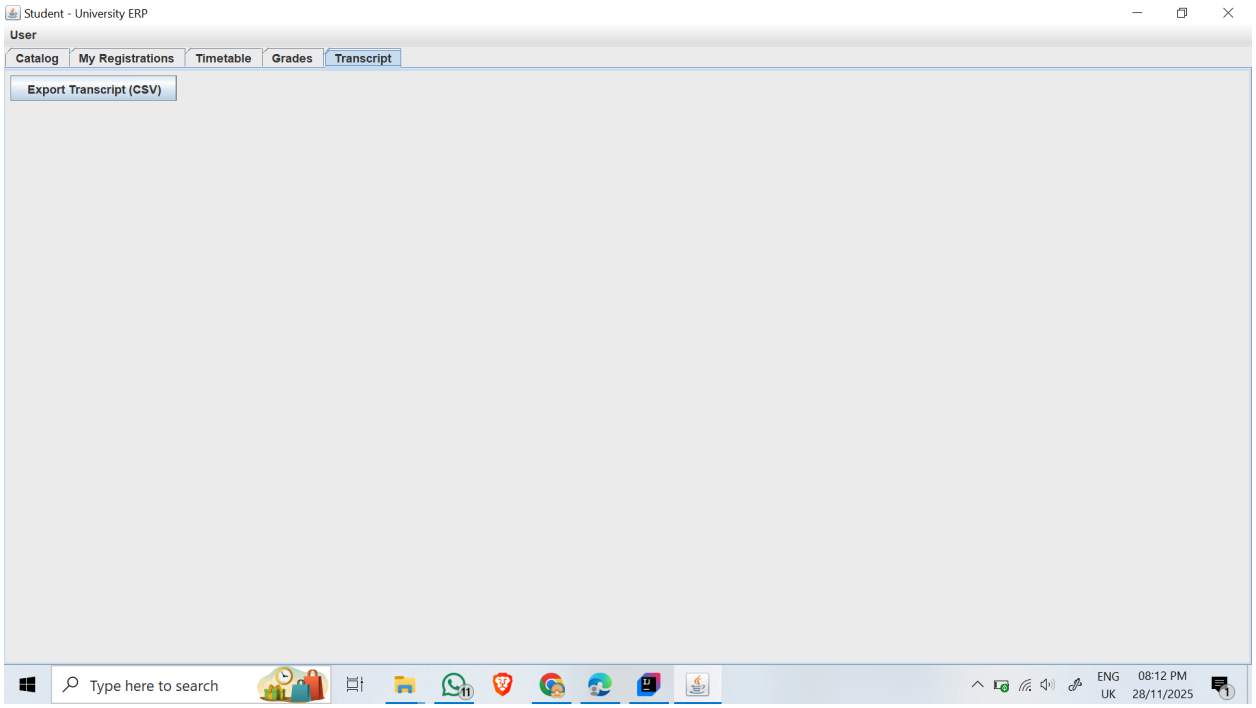

ERP Version control Current File
Project
Main.java
ERP C:\Users\DELL\Onedr
.idea
out
src
edu.univ.erp
access
auth
data
domain
service
ui
util
Main
Main.class
auth.csv
package edu.univ.erp;
import edu.univ.erp.auth.AuthService;
import edu.univ.erp.auth.Session;
import edu.univ.erp.domain.*;
import edu.univ.erp.data.*;
import javax.swin
import java.util.
public class Main
public static
CourseData.init();
Login - University ERP
Message
Welcome stu1 (student)
OK
Run Main
"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=4
ERP > src > edu > univ > erp > Main
10:1 CRLF UTF-8 4 spaces
Type here to search
ENG UK 08:11 PM 28/11/2025

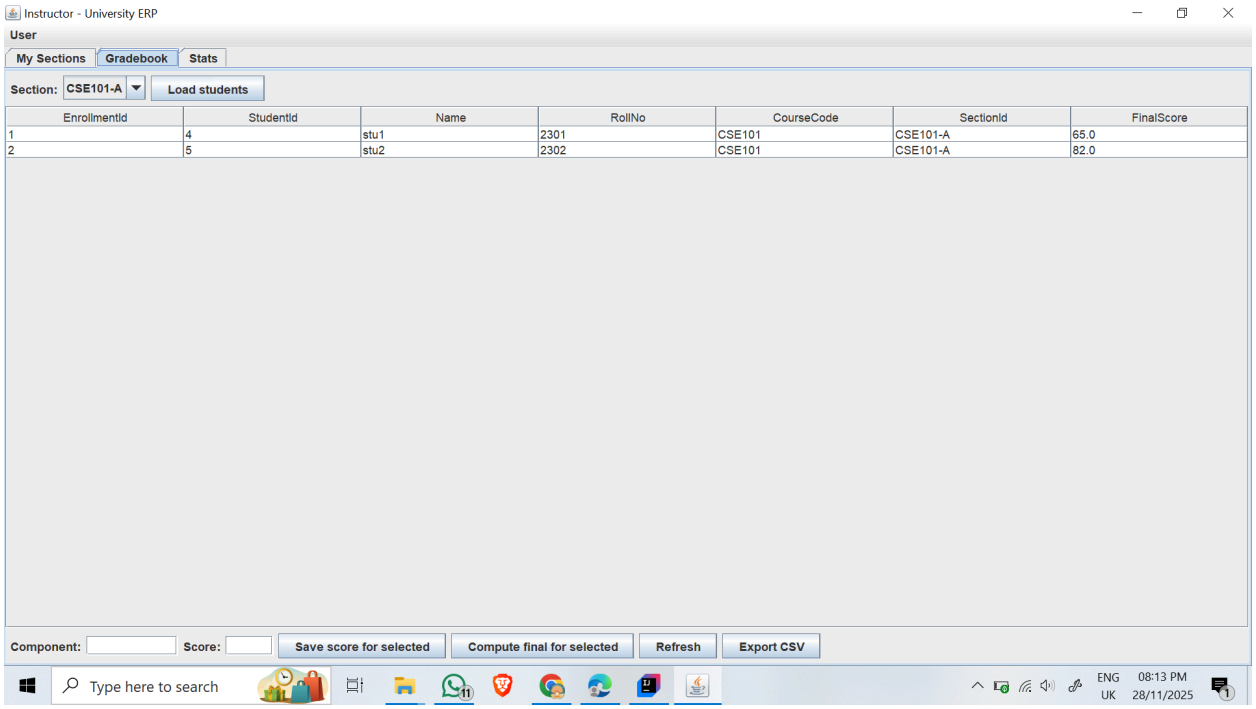


SectionId	CourseCode	Title	Credits	Capacity	Instructor	Day/Time	Room
CSE101-A	CSE101	introduction to programming	4	2	inst1	Mon/Wed 1:00-3:00	C102
CSE102-A	CSE102	Data Structures & Algorithms	4	2	inst2	Mon/Wed 11-12	C102
COM301-A	COM301	Technical Communication	2	150	inst3	Mon/Wed 1-2:30pm	A006



SectionId	CourseCode	Day/Time	Room
CSE101-A	CSE101	Mon/Wed 1:00-3:00	C102
CSE102-A	CSE102	Mon/Wed 11-12	C102


Student - University ERP
User
Catalog
My Registrations
Timetable
Grades
Transcript
Export Transcript (CSV)
Type here to search
ENG
UK
08:12 PM
28/11/2025


Instructor - University ERP
User
My Sections
Gradebook
Stats
Section:
CSE101-A
Load students
EnrollmentId | StudentId | Name | RollNo | CourseCode | SectionId | FinalScore
1 | 4 | stu1 | 2301 | CSE101 | CSE101-A | 65.0
2 | 5 | stu2 | 2302 | CSE101 | CSE101-A | 82.0
Component:
Score:
Save score for selected
Compute final for selected
Refresh
Export CSV
Type here to search
ENG
UK
08:13 PM
28/11/2025

The End....